

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

4 Complexity Theory (mpf23)

complexity class
complementation

- (a) For a complexity class $\mathcal{C}$, let $\text{co-}\mathcal{C} = \{ \overline{L} \mid L \in \mathcal{C} \}$ and say that $\mathcal{C}$ is closed under complementation whenever $\mathcal{C} = \text{co-}\mathcal{C}$.

Argue as to whether the following statements are true, false, or unknown.

- (i) All deterministic time complexity classes are closed under complementation. [3 marks]

- (ii) All non-deterministic time complexity classes are closed under complementation. [3 marks]

Answer:

- (i) True.

Any deterministic Turing machine deciding a language within a time bound can be converted to a deterministic Turing machine that decides its complement within the same time bound; namely, the same machine with the roles ‘yes’ and ‘no’ reversed.

- (ii) Unknown.

In particular, whether or not $\text{NP} = \text{co-NP}$ is an open problem.

NP complexity
class

- (b) For a mapping $f : \Sigma \rightarrow \Sigma$ on an alphabet Σ and a language $L \subseteq \Sigma^*$, define $f[L] = \{ f^{\sharp}(w) \in \Sigma^* \mid w \in L \}$ where $f^{\sharp}(a_1 \cdots a_n) = f(a_1) \cdots f(a_n)$.

Prove that $L \in \text{NP}$ implies $f[L] \in \text{NP}$. [4 marks]

Answer: Let M be a polynomial-time nondeterministic Turing machine recognising L . One may construct a polynomial-time nondeterministic Turing machine $f[M]$ recognising $f[L]$ as follows. The machine $f[M]$ starts by scanning the symbols of its input string, say $a_1, \dots, a_n$, nondeterministically generating a string of the same length, say $b_1 \dots b_n$, while testing that f maps b_i to a_i for all $1 \leq i \leq n$. After this linear nondeterministic preprocessing, $f[M]$ behaves as M on the generated string.

NP-completeness

- (c) Consider the following decision problem.

Q: Given natural numbers m and n in $\mathbb{N}$, and bits $a_{i,j}^{(k)}$ and b_k in $\{0, 1\}$ for $1 \leq k \leq m$ and $1 \leq i, j \leq n$, determine whether the system of equations $\sum_{1 \leq i, j \leq n} a_{i,j}^{(k)} x_i x_j = b_k$ ($1 \leq k \leq m$) with unknowns $x_1, \dots, x_n$ has a solution in arithmetic modulo 2.

- (i) Prove that **Q** is in NP. [3 marks]

Answer: The problem is in NP because one can non-deterministically guess n bits, say $c_1, \dots, c_n$ (respectively standing for values of the unknowns $x_1, \dots, x_n$) and then compute the bit $\sum_{1 \leq i, j \leq n} a_{i,j}^{(k)} c_i c_j$ in arithmetic modulo 2, while checking whether or not it is equal to the bit b_k , in linear time for $1 \leq k \leq m$.

- (ii) By means of a polynomial-time reduction from the problem 3CNF, or otherwise, prove that Q is NP-hard. [*Hint*: Note, for instance, that $x = \neg y$ in the Boolean algebra $\{0, 1\}$ if, and only if, $xx + yy = 1$ in arithmetic modulo 2.] [7 marks]

Answer: Given ϕ in 3CNF construct an instance of Q as follows: introduce unknowns x and x' for every variable x in ϕ and subject them to the equation $xx + x'x' = 1$ (so that solutions in arithmetic modulo 2 force the bit values of x and x' to be complementary). Next, define $\hat{x} = x$ and $\hat{\neg x} = x'$ and for every clause $(\alpha \vee \beta \vee \gamma)$ in ϕ introduce a new variable z and the equations $\hat{\alpha}\hat{\alpha} + \hat{\beta}\hat{\beta} + \hat{\alpha}\hat{\beta} + zz = 0$ (so that $\hat{\alpha} \vee \hat{\beta} = z$) and $zz + \hat{\gamma}\hat{\gamma} + z\hat{\gamma} = 1$ (so that $z \vee \hat{\gamma} = 1$). Argue that: (1) the given transformation is computable in polynomial time, and (2) that a Boolean expression in 3CNF is satisfiable if, and only if, the resulting system of equations has a solution in arithmetic modulo 2.

You may use standard results provided that you state them clearly.